

Received-Power-Aware Carrier Phase Estimator Selection for Turbulent Satellite–Ground FSO Links

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Abstract—Coherent free-space optical links require carrier recovery that remains effective as received power changes with atmospheric turbulence. Data-aided (DA) and non-data-aided (NDA) phase estimators offer complementary operating regions: pilots support phase estimation when the effective signal-to-noise ratio (SNR) is low, while blind estimation preserves every transmitted position for data. This paper develops a two-layer selector that first detects strongly varying received-power windows through a coefficient-of-variation gate and then combines a blind irradiance proxy with a fixed 13 dB effective-SNR decision level. The receiver selects between multi-pilot phase-time least squares and an eighth-power NDA estimator. Simulations of 16-ary amplitude-phase shift keying (16APSK) over additive-noise and five Gamma–Gamma turbulence conditions characterize the complementary estimator behavior. On a separate 29-point selector grid comprising eight AWGN points and seven points in each of the weak, moderate, and strong downlink regimes, the selector recovers the lower-data-BER fixed estimator at 26 points. Relative to fixed NDA at the same data-symbol SNR, representative BER-ratio reductions are 2.3, 2.0, and 1.3 dB in weak, moderate, and strong downlink turbulence, respectively. The results demonstrate that fixed, interpretable statistics can convert the complementary DA/NDA operating regions into repeatable receiver selection across atmospheric conditions.

I. INTRODUCTION

The continued deployment of low-Earth-orbit (LEO) satellite constellations is increasing the capacity required from feeder links between satellites and optical ground stations. Coherent free-space optical (FSO) communication is attractive for this role because coherent reception offers high sensitivity, supports spectrally efficient higher-order modulation, and enables digital compensation of carrier impairments [1], [2]. The atmospheric path nevertheless introduces absorption, scattering, and refractive-index fluctuations. Turbulence produces scintillation in the received irradiance and perturbs the optical phase, while satellite motion and finite laser linewidth add time-varying carrier-frequency and phase errors [2], [3]. These effects make carrier phase recovery (CPR) a central receiver function: the residual phase must be estimated accurately enough to preserve the decision regions of a high-order constellation as the received signal quality changes over time. Recent coherent-FSO studies consequently combine atmospheric mitigation with digital carrier synchronization or robust data-aided processing [2], [4].

Data-aided (DA) and non-data-aided (NDA) carrier estimators provide complementary ways to obtain that phase reference. A DA receiver inserts known pilot symbols, allowing the receiver to form direct phase observations and fit the carrier evolution from regularly spaced references. Data-aided processing has demonstrated stable coherent reception at very low

signal-to-noise ratios (SNRs) in free-space experiments [4]. An NDA receiver instead removes the modulation phase from the received data. The classical Viterbi–Viterbi construction raises phase-shift-keyed samples to the rotational-symmetry order before averaging their phases [5]; the same principle extends to amplitude-phase-shift-keyed constellations by using the number of phase points on each ring [6]. NDA therefore retains every transmitted position for payload and can aggregate the full observation window, whereas DA obtains explicit references at a prescribed pilot density. Under block-varying Gamma–Gamma irradiance, the effective SNR seen by successive processing windows changes with the atmospheric gain. The two reference strategies can consequently favor different operating regions, creating a direct motivation for receiver-side selection across these regions.

This paper develops a blockwise DA/NDA selector for a coherent satellite–ground FSO link using (8, 8) 16-ary amplitude-phase shift keying (16APSK). The selector applies one fixed two-layer decision procedure to every 256-sample digital-signal-processing window. First, a pre-calibrated coefficient-of-variation boundary screens the raw received-power samples. For the remaining windows, a blind received-power proxy is converted to an effective SNR and compared with a fixed 13 dB decision level. This information flow uses quantities available before either carrier-recovery branch produces decoded bits and operates with a shared configuration across turbulence regimes. The resulting study makes three contributions. First, it states a reproducible signal, block, pilot, and total-energy contract that distinguishes the 100-symbol atmospheric channel block from the 256-sample receiver window. Second, across a prescribed grid of 29 additive white Gaussian noise (AWGN) and downlink-turbulence operating points, the selector recovers the fixed estimator with the lower data-bit error rate (BER) at 26 points. Third, Fig. 5 quantifies same-SNR BER-ratio reduction relative to fixed NDA, defined by $10 \log_{10}(P_{b,NDA}/P_{b,SW})$. Representative reductions are 2.3 dB at 10 dB SNR in weak turbulence, 2.0 dB at 10 dB SNR in moderate turbulence, and 1.3 dB at 5 dB SNR in strong turbulence. These values directly report the logarithmic BER ratio at common SNRs. Separately, in the predefined high-SNR work region, the NDA estimator shows an equivalent-SNR advantage of up to 1.9 dB over DA after applying the stated pilot-energy coordinate. This second result reports the NDA–DA work-region advantage on the total-energy coordinate.

The remainder of the paper is organized as follows. Section II defines the signal impairments, atmospheric and re-

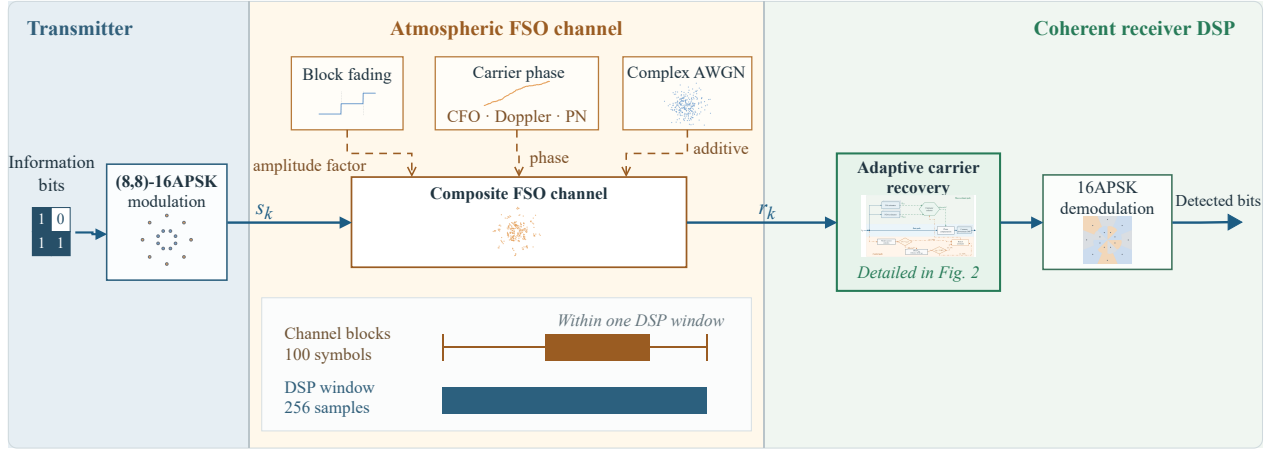


Fig. 1. Coherent satellite-ground FSO link with atmospheric propagation, coherent reception, and the carrier-recovery subsystem expanded in Fig. 2.

ceiver block structure, pilot pattern, and SNR convention. Section III describes the DA and NDA estimators, the two-layer selection statistics, and the fixed decision parameters. Section IV presents the simulation and metric contract, the complementary DA/NDA behavior, and the selector evaluation. Section V concludes the paper.

II. SYSTEM MODEL

A. Signal and impairment model

As shown in Fig. 1, we consider a coherent free-space optical (FSO) link between a satellite and an optical ground station, including both downlink and uplink propagation regimes. The transmitted symbols use (8,8) 16-ary amplitude-phase shift keying (16APSK), with eight equally spaced phase points on each of two amplitude rings, following the modulation definition used by Du *et al.* [6]. The constellation is normalized so that its average symbol energy is $E_s = 1$.

Let $b(k) = \lfloor k/N_{\text{ch}} \rfloor$ identify the atmospheric channel block containing symbol k , where $N_{\text{ch}} = 100$ symbols. The discrete-time complex-baseband observation is

$$r_k = \sqrt{h_{b(k)}} s_k \exp(j\phi_k) + n_k, \quad (1)$$

where s_k is the transmitted symbol, $h_{b(k)}$ is the normalized irradiance, or intensity gain, and $n_k \sim \mathcal{CN}(0, N_0)$ is circular complex Gaussian noise. The square root is required because irradiance scales signal power, whereas the complex-baseband coefficient scales amplitude. Consequently, the instantaneous received signal-to-noise ratio (SNR), before receiver estimation error, is $h_{b(k)}\bar{\gamma}$, where $\bar{\gamma} = E_s/N_0$ is the average pre-fading SNR.

The irradiance follows a Gamma-Gamma model [3] and is constant only within each 100-symbol channel block. It is generated as the product of two unit-mean Gamma variates. The shape-parameter pairs $(\alpha, \beta) = (4.0, 3.0)$, $(2.5, 1.8)$, and $(1.5, 0.8)$ define the weak, moderate, and strong downlink simulation regimes; $(1.2, 0.9)$ and $(1.0, 0.7)$ define the moderate and strong uplink regimes. These parameter pairs define

the numerical fading regimes evaluated below. Treating the atmospheric gain as constant over a short processing interval is consistent with reported FSO timing analyses that place scintillation dynamics on a substantially slower time scale than a microsecond-scale coherent frame [4].

The carrier phase is continuous across channel-block and DSP-window boundaries. It contains a residual frequency offset f_{res} , a linear Doppler rate $\dot{f}$, and laser phase noise:

$$\phi_k = \phi_{\text{init}} + 2\pi f_{\text{res}} k T_s + \pi \dot{f} (k T_s)^2 + \sum_{\ell=0}^k \epsilon_{\ell}, \quad (2)$$

$$\epsilon_{\ell} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 2\pi \Delta\nu T_s),$$

where T_s is the symbol interval, $\Delta\nu = 10$ kHz is the combined laser linewidth, and ϕ_{init} is an arbitrary initial phase. The Wiener-increment variance follows the standard linewidth model used for coherent FSO carrier estimation [6]. Unlike the atmospheric coefficient, this phase is not assumed constant within a channel block.

Equations (1) and (2) keep the impairment roles explicit. The irradiance term controls received signal power, the residual frequency and Doppler-rate terms produce deterministic phase evolution, the Wiener term produces stochastic phase diffusion, and n_k sets the additive-noise floor. In particular, neither the random irradiance nor the carrier phase is absorbed into $\bar{\gamma}$. This convention lets all receiver branches process the same channel realization while differing only in the observations and estimators available to them.

B. Block, pilot, and SNR contract

Two distinct partitions are used. The atmospheric channel block has $N_{\text{ch}} = 100$ symbols and determines where $h_{b(k)}$ is held constant. Carrier recovery and estimator selection operate on nonoverlapping digital signal processing (DSP) windows of $N_{\text{DSP}} = 256$ samples. Because these lengths differ, a DSP window may contain samples from multiple atmospheric channel blocks; the receiver does not assume a single value of

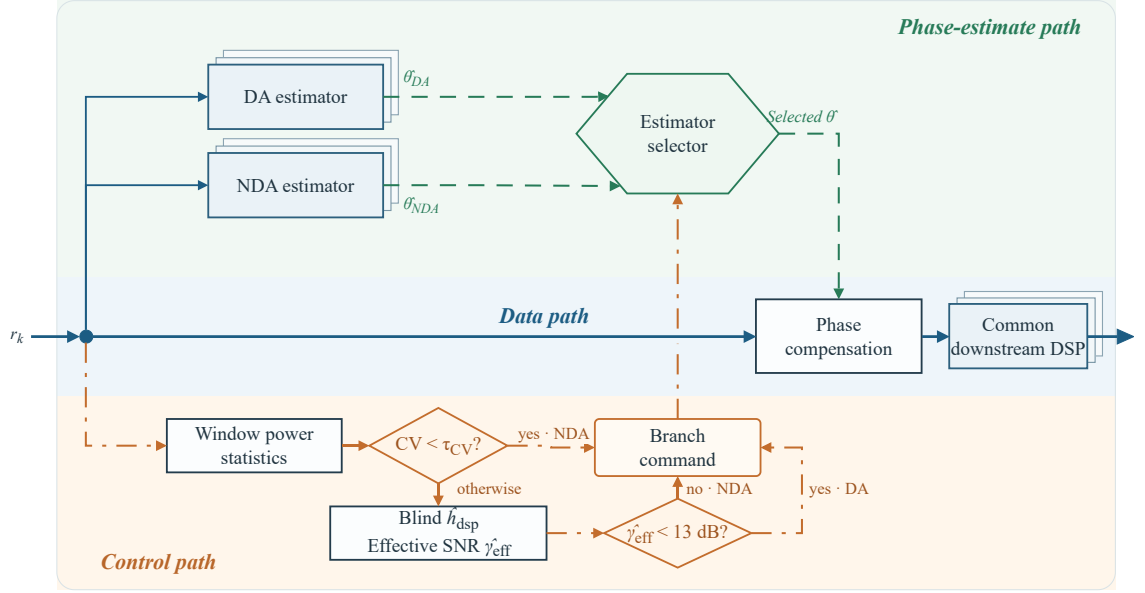


Fig. 2. Carrier-recovery information flow. The received samples feed the DA and NDA phase-estimation candidates and continue along a common data path. The control path summarizes the window statistics and fixed decision configuration used by the two-layer rule in Eqs. (6) and (7); the selected phase estimate enters the shared compensation stage.

h over all 256 samples. Channel-gain estimation is therefore performed on the 100-symbol partition, while phase estimation and method selection use the 256-sample partition.

Within each DSP window, the data-aided layout assigns local indices $0, 4, \dots, 252$ to 64 known pilots and the remaining 192 positions to data. Channel-block boundaries are applied independently of this local layout, so pilots contributing to a channel-gain estimate are selected by their global symbol indices rather than by treating the DSP window as one fading interval. The same transmitted symbols, atmospheric gains, phase sequence, and noise realization are supplied to all compared receiver branches.

For the data-aided branch, one known pilot is placed every $L_p = 4$ transmitted symbols. Thus pilots occupy $\rho_p = 1/4$ of transmitted positions, while the expansion relative to the three data symbols is $1/3$. Comparisons use a total-transmit-energy coordinate: sending three data symbols plus one pilot requires $4/3$ of the energy used by three data-only symbols at the same per-symbol energy. The coordinate offset is therefore

$$\begin{aligned} \rho_p &= \frac{1}{4}, \\ \Delta_p &= 10 \log_{10} \left(\frac{4}{3} \right) = 1.249 \text{ dB}, \\ \gamma_{\text{tot,DA}}^{\text{dB}} &= \gamma_{\text{data}}^{\text{dB}} + \Delta_p. \end{aligned} \quad (3)$$

The non-data-aided branch carries data in every symbol and receives no such coordinate offset. For both branches, bit error rate (BER) is evaluated over information-bearing data positions; pilot positions are excluded from the data-aided denominator. Performance is referenced to the pre-decoding

BER threshold 3.8×10^{-3} , corresponding to 7% overhead hard-decision forward error correction (HD-FEC) [7].

III. METHOD

The receiver uses data-aided (DA) and non-data-aided (NDA) carrier recovery as two complementary processing branches. Both branches operate on a digital-signal-processing window of $N_{\text{DFT}} = 256$ received samples, which is distinct from the 100-symbol atmospheric channel block defined in the system model. A branch is selected once per 256-sample window. The selection statistic is formed directly from the raw received samples, before branch-specific channel equalization and carrier recovery, so that the decision follows the same information flow for every turbulence regime. After selection, the DA path uses pilot-derived channel information, whereas the NDA path uses a blind channel estimate. The carrier-frequency and carrier-phase operations within the two paths are described next.

A. DA and NDA Estimation Stages

The DA branch uses one known pilot every four transmitted symbols. Let n_ℓ denote the index of the ℓ th pilot in the current processing window, p_ℓ its known complex value, and $t_\ell = n_\ell T_s$ its sampling time. Removing the known pilot modulation

and fitting a linear phase model gives

$$\begin{aligned} \vartheta_\ell &= \text{unwrap} \left\{ \angle \left(\frac{r_{n_\ell}}{p_\ell} \right) \right\}, & \vartheta_\ell &\simeq \phi_0 + 2\pi\Delta f t_\ell, \\ \widehat{\Delta f}_{\text{DA}} &= \frac{\sum_\ell (t_\ell - \bar{t})(\vartheta_\ell - \bar{\vartheta})}{2\pi \sum_\ell (t_\ell - \bar{t})^2}, \\ \hat{\phi}_{0,\text{DA}} &= \bar{\vartheta} - 2\pi\widehat{\Delta f}_{\text{DA}}\bar{t}. \end{aligned} \quad (4)$$

Thus, the same set of regularly spaced pilots supplies both the frequency slope and the carrier-phase intercept. The compensated DA samples are obtained by removing $\hat{\phi}_{0,\text{DA}} + 2\pi\widehat{\Delta f}_{\text{DA}}kT_s$ from the received phase. The pilot density and its associated total-energy signal-to-noise ratio (SNR) convention follow the system-model definition; the present section uses that convention without introducing a second SNR coordinate.

With a spacing of four, a 256-sample window contains 64 pilot positions at indices $0, 4, \dots, 252$. The regression treats their unwrapped phases as one ordered sequence rather than as independent single-pilot estimates. Centering the pilot times in Eq. (4) gives the frequency slope from their covariance with phase, while the intercept refers the phase estimate to the beginning of the processing window. The receiver then constructs one phase ramp for all 256 samples. This makes the DA output a carrier-corrected complex sequence with the same length and sample timing as the input window, which allows the downstream demodulator to use the same interface for either selected branch.

The NDA branch removes the (8,8)-16APSK modulation periodicity through the M_0 th-power operation, with $M_0 = 8$. A frequency estimate is first obtained from the spectrum of the raised signal and removed from the received window. Carrier phase is then estimated from the argument of the averaged M_0 th-power samples. For a sample set $\mathcal{I}$, the basic estimate is

$$\hat{\phi}_{\mathcal{I}} = \frac{1}{M_0} \angle \left(\frac{1}{|\mathcal{I}|} \sum_{k \in \mathcal{I}} r_k^{M_0} \right), \quad M_0 = 8. \quad (5)$$

Two windowing modes implement this operation. Under Gamma-Gamma turbulence, $\mathcal{I}$ contains the complete 256-sample processing window and produces one window-level carrier-phase estimate. In the additive white Gaussian noise configuration, the window is divided into $K = 8$ contiguous segments of 32 samples. An M_0 th-power mean-angle estimate is computed at each segment center, the eight raised-domain phases are unwrapped across the window, and linear interpolation produces a sample-wise phase trajectory before division by M_0 . This segmented form follows the intra-window Wiener phase evolution while retaining the same modulation-removal operation as the whole-window form. In both modes, the recovered phase is applied through complex derotation of the corresponding received samples. The DA and NDA branches therefore differ in their available reference information and phase construction, while sharing the same 256-sample decision interval.

The NDA frequency stage precedes these phase modes. Raising the samples to the eighth power removes the ring-

wise angular periodicity and produces a spectral component whose frequency is eight times the carrier-frequency offset. After locating that component, the receiver divides the raised-domain estimate by M_0 , removes the corresponding phase ramp, and applies either the whole-window or segmented phase construction to the frequency-compensated samples. The order is consequently modulation removal, frequency estimation, frequency compensation, and carrier-phase estimation. The resulting derotation preserves the sample order and returns a 256-sample complex window for demodulation.

B. Switching Information Flow and Decision Rule

The switching controller processes the raw received power $q_k = |r_k|^2$ over the current 256-sample window. Its first statistic is the coefficient of variation (CV), which describes the normalized spread of the received power:

$$\begin{aligned} \text{CV} &= \frac{\text{std}(q_k)}{\text{mean}(q_k)}, \\ \tau_{\text{CV}}(\bar{\gamma}_{\text{dB}}) &= 1.10 \left(0.74 + 0.12e^{-\bar{\gamma}_{\text{dB}}/5} \right). \end{aligned} \quad (6)$$

Here, $\bar{\gamma}_{\text{dB}}$ is the nominal data SNR supplied to the receiver, and τ_{CV} is a pre-calibrated SNR-dependent boundary. A window with $\text{CV} < \tau_{\text{CV}}$ is directed to the NDA branch. Otherwise, the controller proceeds to the received-power proxy and effective-SNR stage.

For the second stage, let $\bar{\gamma} = 10^{\bar{\gamma}_{\text{dB}}/10}$ denote the linear nominal data SNR. The controller forms a blind power-based channel proxy, maps it to an effective SNR, and applies the complete branch rule:

$$\begin{aligned} \hat{h}_{\text{dsp}} &= \max \left\{ \frac{1}{N_{\text{DFT}}} \sum_{k=0}^{N_{\text{DFT}}-1} |r_k|^2 - \frac{1}{2\bar{\gamma}}, 10^{-6} \right\}, \\ \hat{\gamma}_{\text{eff,dB}} &= \bar{\gamma}_{\text{dB}} + 10 \log_{10} \hat{h}_{\text{dsp}}, \\ \mathcal{A}(r) &= \begin{cases} \text{NDA}, & \text{CV} < \tau_{\text{CV}}, \\ \text{DA}, & \text{CV} \geq \tau_{\text{CV}} \text{ and } \hat{\gamma}_{\text{eff,dB}} < 13 \text{ dB}, \\ \text{NDA}, & \text{CV} \geq \tau_{\text{CV}} \text{ and } \hat{\gamma}_{\text{eff,dB}} \geq 13 \text{ dB}. \end{cases} \end{aligned} \quad (7)$$

Equation (7) makes the execution order explicit. The CV comparison is evaluated first. Only windows that pass this first boundary use $\hat{h}_{\text{dsp}}$ and $\hat{\gamma}_{\text{eff,dB}}$ for the 13-dB comparison. This arrangement separates windows whose raw power variation matches the pre-calibrated first-stage condition from those for which the effective-SNR decision distinguishes the DA and NDA operating regions.

Operationally, the controller first accumulates q_k across the received window and forms its mean and standard deviation. The normalized spread is compared with τ_{CV} ; an NDA decision at this stage completes the route for that window. For the remaining windows, the already accumulated mean power is reused in $\hat{h}_{\text{dsp}}$, combined with the nominal data SNR, and transformed into $\hat{\gamma}_{\text{eff,dB}}$. The final comparison sends values below 13 dB to DA and values at or above 13 dB to NDA. This staged data flow implements the piecewise rule directly and keeps the first- and second-stage decision variables distinct.

C. Online Variables and Fixed Decision Parameters

The controller separates quantities evaluated for every received window from quantities fixed before operation. The online sample set consists of the 256 raw complex samples r_k . Their squared magnitudes provide the sample mean and standard deviation used by the CV stage, and the same mean power enters $\hat{h}_{\text{dsp}}$. The nominal data SNR $\bar{\gamma}_{\text{dB}}$ is an input to the controller: it selects the value of the CV boundary, determines the noise-subtraction term in the blind power proxy, and provides the reference level for $\hat{\gamma}_{\text{eff,dB}}$. No recovered data symbols or branch-output bit errors enter the selection rule.

The CV coefficients 0.74, 0.12, and 5, the margin factor 1.10, the proxy floor 10^{-6} , and the 13-dB decision level are pre-calibrated fixed parameters. They remain unchanged while the sample mean, sample standard deviation, blind power proxy, and effective SNR are recomputed for each processing window. This division gives a reproducible receiver interface: raw received power and nominal data SNR are the decision inputs; the CV boundary and the 13-dB level define the stored decision configuration; and the selected branch supplies the carrier-corrected samples to the subsequent demodulator. The implementation consequently realizes the sequence CV screening, blind-power evaluation, effective-SNR construction, and fixed-level branch selection stated in Eqs. (6)–(7).

IV. RESULTS

A. Simulation and Metric Contract

We evaluate the data-aided (DA) and non-data-aided (NDA) carrier-phase-recovery paths over six channel conditions: an additive white Gaussian noise (AWGN) reference, weak, moderate, and strong downlink Gamma–Gamma turbulence, and moderate and strong uplink turbulence. The benchmark sweep uses 30 independent seeds at every operating point. Each seed contains 400 blocks of 256 received samples, or 102400 samples per operating point. The sampled average data-symbol signal-to-noise ratios (SNRs) span 5–20 dB for AWGN and 5–26 dB for all turbulent conditions. These common samples provide the basis for the fixed-estimator comparison. The selector-recovery count uses a separate fixed-configuration 30-seed data layer covering 29 prescribed points: eight AWGN SNRs and seven SNRs in each of the three downlink regimes.

Figure 3 also displays additional high-SNR samples so that the low-BER portions of the curves remain visible. Those extension samples use five independent seeds and extend the AWGN, weak, moderate, and strong/uplink curves to 30, 40, 46, and 50 dB, respectively. Together, the 30-seed benchmark and five-seed extension provide the complete displayed SNR range. Figure 4 uses the same merged curve set, while the selector count is evaluated on its separate fixed-configuration 30-seed grid.

The primary metric is the bit error rate (BER), P_b . For NDA, all 256 16-APSK symbols carry data, giving 1024 data bits per block. With one pilot every four symbols, DA is evaluated over the remaining 192 data symbols, giving 768 data bits per

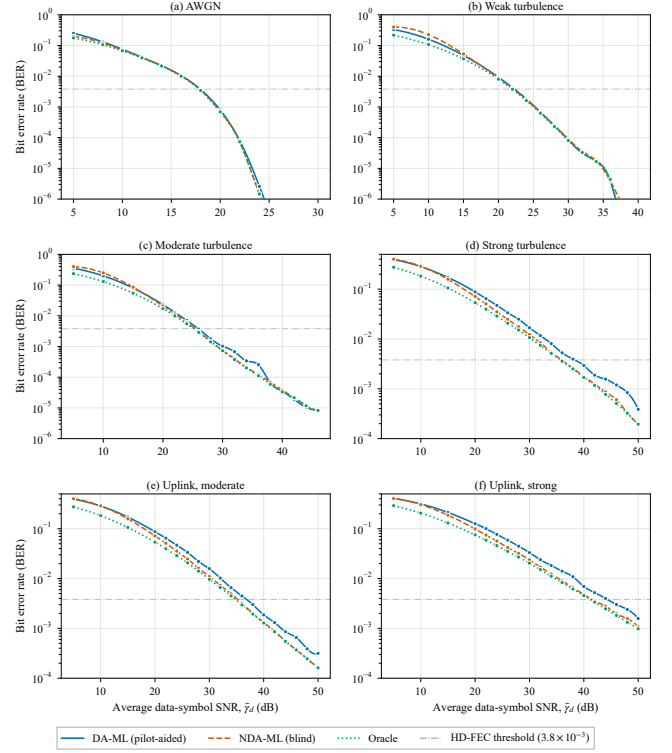


Fig. 3. BER versus average data-symbol SNR for DA-ML, NDA-ML, and the oracle reference under AWGN and five Gamma–Gamma turbulence conditions. Markers denote simulated samples and connecting curves are visual guides. Samples within 5–20 dB (AWGN) and 5–26 dB (turbulence) use 30 seeds with 400 blocks of 256 samples per seed; the high-SNR extension samples use five seeds. The horizontal dash-dotted line marks the 7% HD-FEC reference, $P_b = 3.8 \times 10^{-3}$.

block. This data-BER convention evaluates the decoded payload positions of each branch. The 7% hard-decision forward-error-correction (HD-FEC) reference is $P_b = 3.8 \times 10^{-3}$ [7]; Fig. 3 uses it as a visual performance reference, while the selector statistics follow the data-BER definition.

B. DA–NDA Behavior

Figure 3 shows the complementary operating regions of the two estimators. DA uses known pilot symbols to support carrier estimation, whereas NDA preserves all symbol positions for payload. In the predefined work region $\gamma_{\text{tot}} \geq 15$ dB, the NDA-versus-DA equivalent-SNR advantage is approximately 1.3 dB for strong downlink turbulence, 1.2 dB for moderate uplink turbulence, and 1.9 dB for strong uplink turbulence. These three values are averages over that work region after expressing the two estimators on the stated total-energy coordinate, which includes the $10 \log_{10}(4/3) = 1.249387$ dB pilot-energy offset. They report the NDA–DA work-region comparison on a common total-energy coordinate.

The intersections of the DA and NDA data-BER curves provide a second view of this complementarity. Using linear interpolation in log-BER between simulated samples, the observed crossover SNRs are 18.0, 16.9, and 10.7 dB for weak, moderate, and strong downlink turbulence, respectively. Fig-

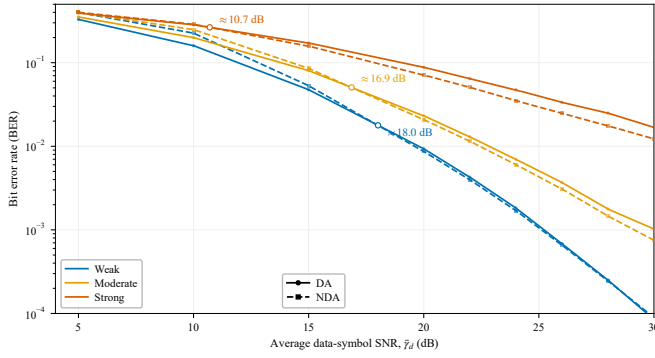


Fig. 4. Observed DA–NDA data-BER crossovers for weak, moderate, and strong downlink turbulence. Solid curves with circular samples denote DA, dashed curves with square samples denote NDA, and open circles mark log-BER-interpolated intersections at 18.0, 16.9, and 10.7 dB. The selector applies its shared fixed 13 dB effective-SNR decision level across these regimes.

ure 4 marks these post-simulation curve observations directly. The receiver uses one fixed 13 dB effective-SNR decision level shared by all evaluated regimes.

C. Switching Evaluation

We quantify the selector relative to a receiver that always uses NDA at the same average data-symbol SNR. For each operating point, the BER-ratio reduction is

$$G_{\text{BER}} = 10 \log_{10} \left(\frac{P_{b,\text{NDA}}}{P_{b,\text{SW}}} \right), \quad (8)$$

where $P_{b,\text{NDA}}$ and $P_{b,\text{SW}}$ are the fixed-NDA and switched-path BERs, respectively. A positive G_{BER} gives the logarithmic BER reduction at a common data-symbol SNR. Figure 5 reports this metric over the three downlink turbulence regimes. Representative positive values are 2.3 dB at 10 dB in weak turbulence, 2.0 dB at 10 dB in moderate turbulence, and 1.3 dB at 5 dB in strong turbulence. These anchors illustrate the reduction obtained when the selector activates the branch favored by its fixed decision rule.

Across the selector grid, 29 operating points are evaluated: eight AWGN SNRs plus seven SNRs in each of the weak, moderate, and strong downlink regimes, i.e., $29 = 8 + 7 + 7 + 7$. Using the standard data-BER definition to identify the lower-BER fixed estimator, the selector recovers that estimator at 26 of the 29 points. This count and Fig. 5 describe complementary aspects of operation: the count measures branch recovery across the prescribed grid, whereas G_{BER} measures the BER reduction relative to fixed NDA at each common SNR. Together, they show that a single fixed 13 dB decision level shared across regimes converts the DA–NDA operating-region complementarity into repeatable estimator selection.

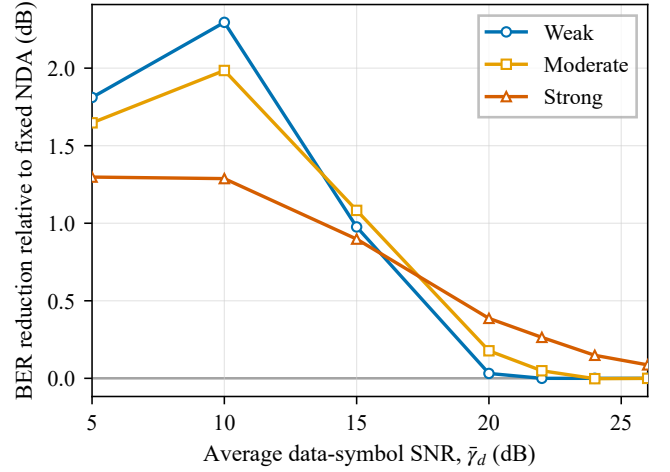


Fig. 5. Same-SNR BER reduction of the switched path relative to fixed NDA. $G_{\text{BER}} = 10 \log_{10}(P_{b,\text{NDA}}/P_{b,\text{SW}})$, for weak, moderate, and strong downlink turbulence. Positive values indicate a lower switched-path BER at the same average data-symbol SNR and directly report the logarithmic BER ratio.

V. CONCLUSION

This paper presented a two-layer DA/NDA estimator selector for carrier phase recovery in turbulent satellite–ground FSO links. A coefficient-of-variation gate identifies windows with pronounced power variation, after which a blind irradiance proxy and a fixed 13 dB effective-SNR level select between multi-pilot phase-time least squares and eighth-power NDA estimation. The design preserves the distinct strengths of the two estimators while keeping the online decision variables directly connected to received-window statistics. Across the prescribed AWGN and downlink evaluation grid, the selector recovers the lower-data-BER fixed estimator at 26 of 29 operating points. At representative same-SNR points, it reduces BER relative to fixed NDA by 2.3 dB in weak downlink turbulence, 2.0 dB in moderate downlink turbulence, and 1.3 dB in strong downlink turbulence when expressed as $10 \log_{10}(P_{b,\text{NDA}}/P_{b,\text{SW}})$. Together with the observed DA–NDA crossovers and the NDA work-region advantages under stronger fading, these results establish a compact, interpretable path for matching the carrier phase estimator to changing atmospheric conditions.

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